

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

ACADEMIC YEAR 2025/2026

OCTOBER EXAMINATION

AMIT3353 MOBILE APPLICATION DEVELOPMENT

FRIDAY, 10 OCTOBER 2025

TIME: 3.00 PM – 5.00 PM (2 HOURS)

DIPLOMA IN COMPUTER SCIENCE

Instructions to Candidates:

Answer **ALL** questions. All questions carry equal marks.

AMIT3353 MOBILE APPLICATION DEVELOPMENT

The MyGOV Malaysia mobile application is a national digital initiative aimed at streamlining access to government services through a single, unified platform. Launched in mid-2025, the app was developed as part of Malaysia's digital transformation efforts under the Malaysia Digital Economy Blueprint (MyDIGITAL). It brings together essential public services from various ministries and agencies into one mobile interface, reducing the need for physical visits and improving service delivery for all Malaysians.

MyGOV Malaysia is integrated with MyDigital ID, allowing secure and convenient authentication across different government services. This secure digital identity enables users to sign in once and access multiple services without repeatedly entering their credentials. The app promotes inclusivity by offering multilingual support and 24/7 access to essential services, making it especially valuable for users in rural and underserved areas.

The app includes a range of key features and functionalities designed to improve service access, simplify navigation, and enhance the user experience:

- ❖ **All-in-one service access:** MyGOV Malaysia brings together over 100 government services into a single mobile app. Users can perform tasks such as renewing road tax, filing income tax, applying for healthcare support, or checking eligibility for government cash aid. This saves time and reduces the need to visit multiple websites or physical offices.
- ❖ **MyDigital ID integration:** The app is linked with Malaysia's official Digital ID system, allowing users to log in securely. With single sign-on, users only need one account to access all available services. This improves both convenience and data protection.
- ❖ **Real-time application status tracking:** Users can track the status of their applications directly in the app. For example, they can check if a license renewal is approved or if additional documents are needed. Notifications will be sent automatically to inform users of any updates.
- ❖ **Built-in ePayment integration:** The app allows users to make payments for government services using FPX or Touch 'n Go eWallet. This includes paying for fines, license renewals, bills, and other public charges. Transactions are fast, secure, and can be done fully within the app.
- ❖ **User-friendly navigation:** The app has a clean and user-friendly design. Services are grouped into clear categories such as Health, Transport, Education, and Welfare, making it easy to browse. A search bar is also available to help users quickly find what they need.
- ❖ **Multi-language and inclusive access:** MyGOV Malaysia currently supports Bahasa Malaysia and English, making it accessible to most users. It is available for use at any time of the day, so users can manage their tasks whenever it is most convenient. This is especially helpful for people living in rural or remote areas.

References:

Digital Government Malaysia (2025). *MyGOV Malaysia Mobile App to Be Launched Next Month*. Retrieved from: <https://www.digital.gov.my/en-GB/siaran/Aplikasi-Mudah-Alih-MyGOV-Malaysia-Bakal-Dilancarkan-Bulan-Depan>

OpenGov Asia (2025). *Malaysia: MyGOV Mobile App to Enhance Public Service Delivery*. Retrieved from: <https://opengovasia.com/malaysia-mygov-mobile-app-to-enhance-public-service-delivery/?c=my>

Answer the following questions based on the above-mentioned scenario.

AMIT3353 MOBILE APPLICATION DEVELOPMENT**Question 1**

- a) A mobile operating system (OS) is the software that allows smartphones, tablets, and other mobile devices to run applications and programs.
- (i) Examine **TWO (2)** types of mobile OS that support the MyGOV Malaysia mobile app on mobile devices. (6 marks)
- (ii) Justify why these two mobile OS help the MyGOV Malaysia mobile app run effectively on mobile devices. (4 marks)
- b) In Jetpack Compose, delivering a smooth UI experience is vital for maintaining user satisfaction, encouraging engagement, and contributing to the success of a mobile application.
- Adapt **THREE (3)** methods that can be used to optimise layout performance. (15 marks)

[Total: 25 marks]

Question 2

- a) Malaysia is home to citizens from diverse racial backgrounds who use various types of mobile devices.
- Apply **TWO (2)** appropriate methods that the MyGOV Malaysia mobile app can use to serve users of different races and support the wide variety of devices available in the market. (10 marks)
- b) Demonstrate how the MyGOV Malaysia mobile app notifies users in Malaysia when a new update is available. (3 marks)
- c) Identify the suitable intent type (**explicit** or **implicit**) for each task in the MyGOV Malaysia mobile app, with a simple justification.
- (i) Navigating from the home screen to the settings screen. (3 marks)
- (ii) Sending an email to government support. (3 marks)
- (iii) Creating a new request for driving license renewal. (3 marks)
- (iv) Opening a map to navigate to the nearest Urban Transformation Centre (UTC). (3 marks)

[Total: 25 marks]

AMIT3353 MOBILE APPLICATION DEVELOPMENT**Question 3**

- a) Mobile applications communicate with servers using various protocols and methods. The most common approach involves using HTTP or HTTPS requests to send and receive data, often in formats like JSON or XML. Other methods include using WebSockets for persistent, real-time communication or relying on push notifications from a server.
- (i) Define the terms *Front End* and *Back End*. (4 marks)
- (ii) Illustrate mobile system architecture using a diagram. (5 marks)
- b) Phone storage is used to store files such as photos, videos, applications, and other data.
- (i) Differentiate **THREE (3)** differences between internal and external storage in terms of removability, availability, and accessibility in a table format. (6 marks)
- (ii) Provide **ONE (1)** local storage method for both internal and external storage, along with a content example from the MyGOV Malaysia mobile app. (4 marks)
- c) Classify the following uses in the MyGOV Malaysia mobile app under the correct location permission (**foreground** or **background**), with a simple explanation.
- (i) Get direction to the nearest Urban Transformation Centre (UTC). (3 marks)
- (ii) Alerts for nearby government events. (3 marks)

[Total: 25 marks]

Question 4

- a) Relate the applications of geocoding to their relevance and benefits for users of the MyGOV Malaysia mobile app. (6 marks)
- b) Mobile phone cameras are versatile tools used by a wide range of applications, from basic photography to advanced features like augmented reality and video conferencing.
- (i) Describe **TWO (2)** ways a mobile application can handle a camera request when access to the camera is required. (6 marks)
- (ii) Determine the most suitable way for the MyGOV Malaysia mobile app to handle a camera request, with appropriate justification. (3 marks)
- c) Mobile app packaging is the process of preparing an application for distribution and deployment.
- (i) Analyse **ONE (1)** distribution channel used by the government to deliver the MyGOV Malaysia mobile app, with the reasons for the choice. (5 marks)
- (ii) Explain a suitable monetisation model for the MyGOV Malaysia mobile app. (5 marks)

[Total: 25 marks]